

Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(janitor)
library(readxl)
library(snakecase)
library(scales)

# file organization
library(here)

# plots
library(patchwork)
library(viridis)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "Aquatic Insects")
```

```
# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                          sheet = "Water Quality",
                          na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning data

```
# creating a new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sampling
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# creating a new object
water_quality_clean <- water_quality |>
  # cleaning column names
  clean_names() |>
  # use a filtering join, only include sites in ncos_sites
  semi_join(ncos_sites,
            by = "site") |>
  # creating new column to join with later data frames
  unite("date_site", date, site,
        remove = TRUE) |>
  # grouping by date/site
  group_by(date_site) |>
  # calculating median, pH, salinity, and DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungrouping data frame
  ungroup() |>
  # filtering out outliers
  filter(date_site != "2022-08-12_NVBR")
```

```
# create a new object from taxon_list
taxon_list_clean <- taxon_list |>
  # change names in verbatim_name to snake case
```

```
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))
```

```
# create a new object from aq_ins
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filter join to only include sites matching with ncos_sites
semi_join(ncos_sites,
          by = "site") |>
# rename column date_on_vial to date
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include FB250/FB 250 samples
filter(sample_type %in% c("FB250", "FB 250")) |>
# pivot data frame
pivot_longer(
  # columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for the old column names
  names_to = "taxon",
  # new column for the values in each of the old columns
  values_to = "abundance"
) |>
# group by date, site, and taxon
group_by(date, site, taxon) |>
# summarize to find average abundance per liter
summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
# ungroup
ungroup() |>
# combine date and site into date_site, don't remove date and site columns
unite("date_site", date, site, remove = FALSE) |>
```

```

# left join water_quality_clean by date_site column
left_join(water_quality_clean, by = "date_site") |>
# left join taxon_list_clean by taxon(aq_ins)/verbatim_name(taxon_list) column
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# mutate to create a column for full names of site codes
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelp Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
))

```

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

```

Visualizations from class

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,

```

```

        color = "red",
        size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
      y = "Median salinity (ppt)",
      title = "Site salinity") +
# different theme
theme_bw()

```

```

# base layer
copepods <- ggplot(data = copepods,
                  # x-axis: site
                  # y-axis: mean abundance/L
                  mapping = aes(x = site_full,
                                y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Copepods") +
# different theme
theme_bw()

```

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

In part 1d, you should end up with a three panel figure. All three panels should have site on the x-axis, but each panel will have a different y-axis.

In part 1c, you will create one component of the three panel figure. Provide a list of the axes and geometries you will need to create a figure in 1c showing differences between sites in ostracod abundance.

- x-axis: `site_full`
- y-axis: `log_ave_lit`
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary`

b. Wrangle your data

The `aq_ins_clean` data frame will need some further wrangling before you create the figure you described in part 1a.

In this section, create the data frame you will need to make your figure.

At the end of your code chunk, display 10 random rows from your data frame using `slice_sample()`.

```
# create a new object from aq_ins_clean
ostracods <- aq_ins_clean |>
  # filter to only include ostracod taxon
  filter(taxon == "ostracod") |>
  # log transform average abundance/L as a new column
  mutate(log_ave_lit = log(ave_lit + 1))

# display 10 rows from ostracods
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
  date_site      date      site  taxon  ave_lit med_ph med_sal med_do
  <chr>         <dtm>         <chr> <chr>   <dbl>   <dbl>   <dbl>   <dbl>
1 2022-01-24_NEC 2022-01-24 00:00:00 NEC  ostra~  0.267   NA      NA      NA
2 2023-02-12_NEC 2023-02-12 00:00:00 NEC  ostra~  2.13    NA      NA      NA
3 2022-05-11_NDC 2022-05-11 00:00:00 NDC  ostra~  0        9.11    0.398    9.4
4 2022-05-05_NEC 2022-05-05 00:00:00 NEC  ostra~  0.267    9.45   63.8     7.3
5 2023-05-04_NEC 2023-05-04 00:00:00 NEC  ostra~  0.667    8.05    NA      8.55
6 2023-08-12_NEC 2023-08-12 00:00:00 NEC  ostra~  0        NA      NA      NA
7 2021-11-12_NPB 2021-11-12 00:00:00 NPB  ostra~  79.2    NA      NA      NA
8 2021-11-16_NPB 2021-11-16 00:00:00 NPB  ostra~  72.1     7.3    2.38   12.1
9 2023-07-17_NPB 2023-07-17 00:00:00 NPB  ostra~  4.53    NA      NA      NA
10 2022-08-12_NEC 2022-08-12 00:00:00 NEC  ostra~  0        7.37    NA      NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
```

```
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <chr>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer
ostracods <- ggplot(data = ostracods,
                    # x-axis: site
                    # y-axis: mean abundance/L
                    mapping = aes(x = site_full,
                                  y = log_ave_lit)) +

# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +

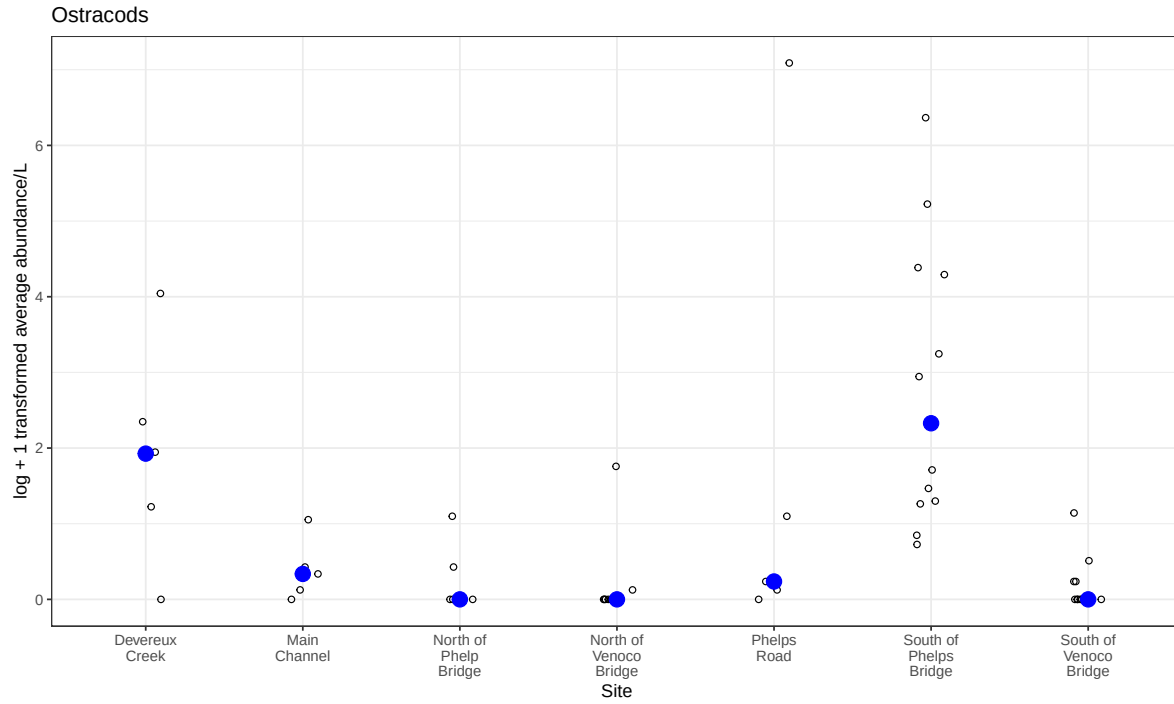
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +

# wrapping x-axis text
scale_x_discrete(labels = label_wrap(10)) +

# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +

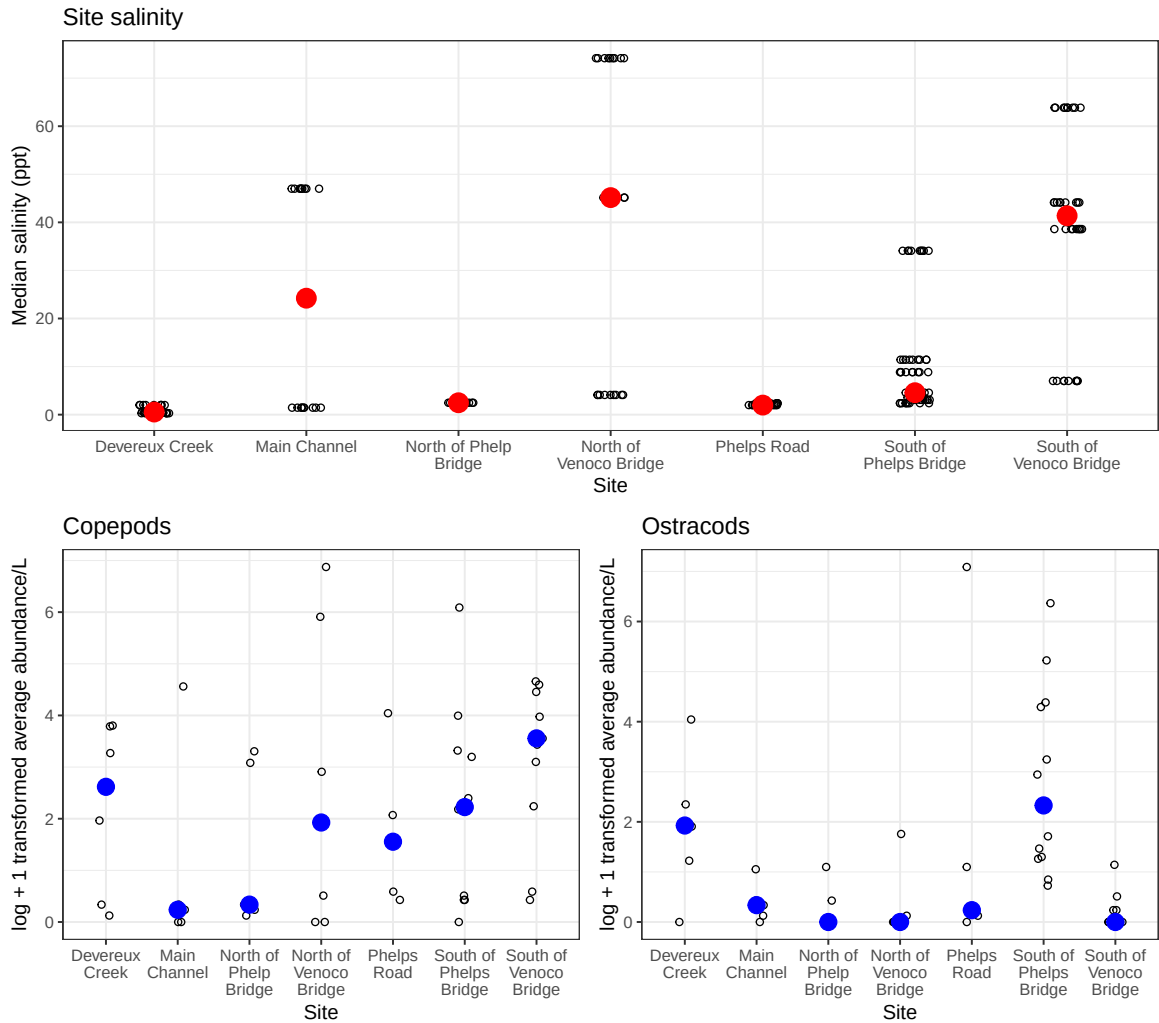
# different theme
theme_bw()

# displaying object
ostracods
```



d. Create a multipanel figure

```
# stack salinity on top and copepods and ostracods side by side
salinity / (copepods + ostracods)
```



e. Interpret your figure

In 1-2 sentences each, describe:

- differences between sites in copepod and ostracod abundance
 - On average, there appears to be a higher abundance of copepods across sites compared to ostracods. We can visually tell because of how the blue points representing the central tendency of abundance are generally higher in the y-axis direction in the copepod plot than in the ostracod plot.
- how ostracod abundance may relate (or not) to site-level differences in salinity

- There doesn't appear to be a clear relationship between ostracod abundance and salinity, which we can visually tell because the positions of the median points for ostracod abundance don't appear any higher or lower in response to site salinity (they don't consistently "move" with each other). We might otherwise expect an inverse correlation, but sites like North of Phelps and Phelps Road disrupt this idea.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

At the end of part 2c, you should have a figure:

- depicting log-transformed average abundance/L on the y-axis and salinity on the x-axis
- with different panels for each taxon in `aq_ins_clean` with free scales for both axes
- with each taxon in its own color

Provide a list of values, geometries, and/or arguments that would be appropriate to make a figure matching this description.

- x-axis: `med_sal`
- y-axis: `log_ave_lit`
- color: `taxon`
- geometry: `geom_point()`
- facets: `taxon`

b. Wrangle your data

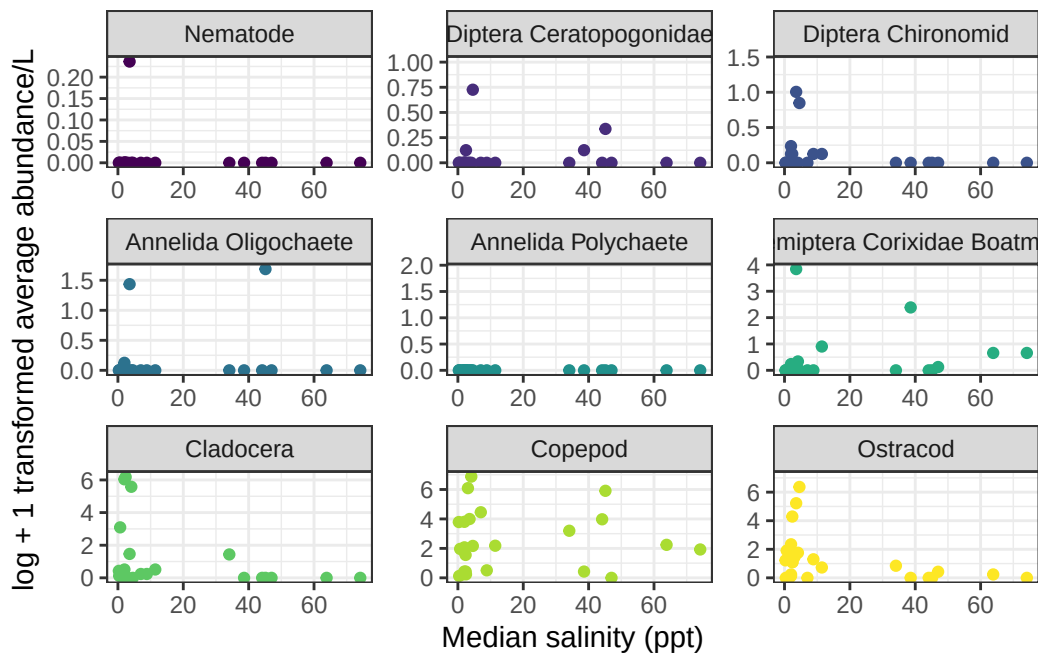
```
# create a new object from aq_ins_clean
aq_ins_salinity <- aq_ins_clean |>
  # log transform average abundance/L as a new column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # set taxon to title case and reorder by max abundance
  mutate(taxon_full = to_any_case(taxon, case = "title"),
         taxon_full = as_vector(taxon_full),
         taxon_full = fct_reorder(taxon_full, log_ave_lit, max)) |>
  # arrange data frame by taxon
  arrange(desc(taxon_full)) |>
  # select columns of interest
  select(med_sal, log_ave_lit, taxon_full)
```

```
# display 10 rows from aq_ins_salinity
slice_sample(aq_ins_salinity, n = 10)
```

```
# A tibble: 10 x 3
  med_sal log_ave_lit taxon_full
  <dbl>     <dbl> <fct>
1 NA         3.31 Copepod
2 NA         0.236 Ostracod
3 NA         0 Hemiptera Corixidae Boatman
4 NA         0 Annelida Oligochaete
5 0.398      0 Annelida Polychaete
6 NA         3.97 Cladocera
7 NA         0 Annelida Oligochaete
8 63.8       0.236 Ostracod
9 NA         0 Nematode
10 8.84      0 Hemiptera Corixidae Boatman
```

c. Code your figure

```
# base layer: ggplot
ggplot(data = aq_ins_salinity,
  # set x- and y-axes, differentiate color by taxon
  aes(x = med_sal,
    y = log_ave_lit,
    color = taxon_full)) +
# first layer: point geometry
geom_point() +
# create panels by taxon, set free scale y-axis
facet_wrap(~ taxon_full,
  scale = "free") +
# add labels to axes
labs(x = "Median salinity (ppt)",
  y = "log + 1 transformed average abundance/L") +
# change colors
scale_color_manual(values = viridis(9)) +
# change theme
theme_bw() +
# remove legend
theme(legend.position = "none")
```



d. Interpret your figure

Across all taxa, there appears to be a general negative relationship between average abundance and salinity. On average, as we move along the x-axis, the points tend to be positioned lower along the y-axis. This relationship is weaker with taxa like the Copepod and Annelida Polychaete, and stronger with taxa like the Ostracod and Cladocera.